

Sooyong Shin

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Interest

I am interested in **Computer Vision** with a focus on **Robot Perception** and high-level scene understanding. I am also open to exploring how **Multimodal Models** and **Vision–Language Integration** can bridge the gap between visual intelligence and actionable robotic intelligence, aiming to facilitate more intuitive and context-aware interactions in dynamic environments.

Keywords: Robot Perception, Vision–Language Models, Vision–Language–Action Model

Education

Korea Advanced Institute of Science and Technology <i>Bachelor of Science, Electrical Engineering, and Computer Science</i>	<i>Daejeon, South Korea</i> <i>Feb 2018 – Feb 2027</i>
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Research Experience

NMAIL Lab (Supervisor: Sungho Jo) <i>Research Intern</i> ◦ Research on Vision Perception & Vision–Language–Action models	<i>KAIST, Daejeon, South Korea</i> <i>March 2026 – Present</i>
Urban Robotics Lab (Supervisor: Hyun MYUNG) <i>Research Intern</i> ◦ Research on Robot Perception	<i>KAIST, Daejeon, South Korea</i> <i>June 2025 – Feb 2026</i>
<i>Individual Study</i> ◦ Studying ROS and Visual SLAM	<i>Mar 2025 – June 2025</i>
SIIT Lab (Supervisor: Junmo Kim) <i>Individual Study</i> ◦ Studying Vision Language Multimodal	<i>KAIST, Daejeon, South Korea</i> <i>Jul 2024 – Aug 2024</i>

Work Experience

KC-ML2 <i>Research Intern</i> ◦ Research on Relocalization for Robot Mapping ◦ Focused on Text-based Failure Recovery in Monocular Visual SLAM ◦ Manuscript in preparation. Co-first author work conducted during the internship at KC-ML2 (2024–2025). ◦ Skills: C++, Python, ORB SLAM, YOLO, paddle OCR	<i>Seoul, South Korea</i> <i>Aug 2024 – Feb 2025</i>
Bookend <i>Software Engineer</i> ◦ Developed Sentensify 🔗 (chrome webstore 🔗), a multilingual AI writing and paraphrasing tool leveraging various Large Language Models ◦ Role: Full-Stack Engineer ◦ Skills: JavaScript(React), Python(FastAPI), MongoDB, ChatGPT API	<i>Daejeon, South Korea</i> <i>May 2023 – Aug 2024</i>
<i>Software Engineering Intern</i> ◦ Developed TYPHERE , a writer-publisher mobile platform designed to streamline and enhance the book publishing process ◦ Role: Front-End Engineer ◦ Skills: JavaScript(React-Native)	<i>Dec 2022 – Feb 2023</i>

Projects

KAIST Swimming Team KAORI Webpage

Since 2024

- Developed a webpage to archive swimming records and member information, enabling analysis of performance trends and data distribution (www.kaist-kaori.org )
- Role: Full-Stack developer
- Skills: JavaScript(React), Python(Fast API), MongoDB

Skills

ML/DL: PyTorch

Robotics: ROS, Visual SLAM (ORB-SLAM)

Programming: Python, C++, JavaScript(TypeScript)

Database: MongoDB, SQL

others: React/React-Native, FastAPI, Jira, Confluence

Leadership Experience

KAIST MR(Microbot Research)

member

Daejeon, South Korea

Since Mar 2025

AI Robotics KR

Community Manager

South Korea

Oct 2024 – Sep 2025

Community Lead

Sep 2025 – Present

Republic of Korea Army

Sergeant, honorable discharge

Paju, South Korea

Mar 2019 – Nov 2020

Korean Red Cross Lifeguard

Daejeon, South Korea

Since Aug 2018

KASIT Swimming Team KAORI

Senior member

Daejeon, South Korea

Since Mar 2018

President of KAORI

2022

Chairman of KAIST Swimming Competition

2023